

Paper ID : 61028
BCA (Semester-I) Examination, 2022
(Session : 2021-24)
COMPUTER APPLICATION
[Paper Code : BCA-101]
(Mathematical Foundation)

Time : Three Hours

[Maximum Marks : 50]

Note: Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any five questions.

1. (a) For what value of x , the matrix

$$\begin{bmatrix} 3-x & 2 & 2 \\ 2 & 4-x & 1 \\ -2 & -4 & -1-x \end{bmatrix} \text{ is singular?}$$

(b) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, find the value of $A^2 - 4A + 3I$, where I is the unit matrix with proper dimension.

2. (i) If $y = \sin(m \sin^{-1} x)$, prove that $(1 - x^2)y'' + m^2y = 0$.

(ii) If $y^{\frac{1}{m}} + y^{\frac{1}{m}} = 2x$, prove that $(x^2 - 1)y'' + xy' - m^2y = 0$.

3. If $y = e^{a \sin^{-1} x}$, prove that $(1 - x^2)y'' - xy' - a^2y = 0$.

4. If $u = \sqrt{x^2 + y^2 + z^2}$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{2}{u}$

5. Evaluate $I = \int \frac{\tan x \cdot \sec^2 x}{1 - \tan^2 x} dx$

6. Evaluate double integral $I = \int_0^2 \int_0^2 (x^2 + y^2) dx dy$

7. Solve the following:

$$(x^2 - y \cdot x^2) dy + (y^2 + x \cdot y^2) dx = 0$$

8. Apply Maclaurin's theorem to obtain the expansion of $\sec x$.

9. Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = 2e^{3x}$

10. Write short notes on the following:

- (a) Transpose matrix
- (b) Singular matrix
- (c) Taylor's theorem
- (d) Inverse matrix

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